3/6/23

About the proof of the JCF main theorem:

Plan to prove

(1) Every $A \in M_n$ is similar to an upper triangular matrix (Schur's Theorem) - done

(2) Show every upper triangular is similar to a block-diagonal matrix with each block being triangular (so eigenvalues are on the diagonal) and diagonal entries of each block are equal.

(3) Show that upper triangular matrices (blocks in step (2)) are similar to direct sums of Jordan blocks.

$$\underset{\in M_n}{[A]} \overset{(1)}{\sim} [T] \overset{(2)}{\sim} \begin{bmatrix} T_1(\lambda_1) & & 0 \\ & \ddots & \\ 0 & & T_k(\lambda_k) \end{bmatrix}$$

$$\overset{(3)}{\sim} \begin{bmatrix} J_{n_1}(\lambda_1) & & & 0 \\ & J_{n_2}(\lambda_2) & & \\ & & \ddots & \\ 0 & & & J_{n_m}(\lambda_m) \end{bmatrix}$$

The biggest challenge is to show uniqueness of the Jordan blocks $J_{n_1}(\lambda_1), \ldots, J_{n_m}(\lambda_m)$.

Let's discuss "uniqueness".

We'll see: The JCF of matrix A is completely determined by knowing

- the eigenvalues of A
- the sizes of the Jordan blocks
- number of Jordan blocks corresp. to each eigenvalue.

<u>Comment</u> The "uniqueness" also implies that every matrix similar to A has the "same" JCF.

Analyze similarity and its effects on JCF:

- Say $A \overset{\text{similar}}{\sim} B$. Then $\forall \lambda \in \mathbb{C}$, $A - \lambda I \sim B - \lambda I$ and so $(A-\lambda I)^m \sim (B - \lambda I)^m$, for $m = 1, 2, \ldots$ $(A, B \in M_n)$

- It suffices to show the Jordan blocks in $J = JCF(A)$ are completely determined by

$$\operatorname{rank}(J - \lambda I)^m, \quad m = 1, 2, \ldots, n$$
$$\text{for } \lambda \in \sigma(A) = \sigma(J)$$
$$\left(\operatorname{rank}(J-\lambda I)^m = \operatorname{rank}(A - \lambda I)^m\right)$$

Here's how:

- Let $J_k(\mu) \in M_k$, $\mu \in \mathbb{C}$, let $m \geq 1$

If $\mu \neq 0 \Rightarrow \operatorname{rank} J_k(\mu)^m - \operatorname{rank} J_k(\mu)^{m+1}$

$$= k - k = 0$$

If $\mu = 0$ & $m \geq k \Rightarrow$

$$\operatorname{rank} J_k(0)^m - \operatorname{rank} J_k(0)^{m+1} =$$

$$= 0 - 0 = 0$$

If $\mu = 0$ & $m < k \Rightarrow$

$$\operatorname{rank} J_k(0)^m - \operatorname{rank} J_k(0)^{m+1} = 1$$

- Now let $J \in M_n$ be a Jordan matrix and $\lambda \in \sigma(J)$. Define

$$r_m(\lambda) := \operatorname{rank}(J - \lambda I)^m, \quad m = 1, 2, \ldots$$

$$r_0(\lambda) = n \quad \text{(by convention)}$$

<u>Then</u> if we let

$$d_m(\lambda) := r_{m-1}(\lambda) - r_m(\lambda)$$

$$= \operatorname{rank}(J - \lambda I)^{m-1} - \operatorname{rank}(J - \lambda I)^m$$

$\overset{!}{=}$ total number of blocks $J_k(\lambda)$ in J of sizes $k \geq m$

(because this difference $d_m(\lambda)$ counts 1 for every block of size $k \geq m$ & 0 for all other blocks)

Finally, the number of Jordan blocks corresp. to λ & of size $k = m$ is equal to

$$d_m(\lambda) - d_{m+1}(\lambda)$$

$$= r_{m-1}(\lambda) - r_m(\lambda) - r_m(\lambda) + r_{m+1}(\lambda)$$

$$= r_{m+1}(\lambda) - 2r_m(\lambda) + r_{m+1}(\lambda)$$

(we can compute all these based on A)

$$r_m(\lambda) = \text{rank}(J-\lambda I)^m = \text{rank}(A-\lambda I)^m$$

("$d_m(\lambda) - d_{m+1}(\lambda)$" Weyl characteristic)

<u>Example</u>

Say $A \sim J = \begin{bmatrix} J_3(2) & & & 0 \\ & J_2(2) & & \\ & & J_2(2) & \\ 0 & & & J_1(2) \end{bmatrix}$

$$J = J_3(2) \oplus J_2(2) \oplus J_2(2) \oplus J_1(2)$$

$$\sigma(J) = \{2,2,2,2,2,2,2,2\}$$

Compute $J - 2I$

$\text{rank} = 2+1+1+0$
$= 4$

$(J-2I)^2$

$\text{rank} = 1+0+0+0$
$= 1$

$(J-2I)^3$

$\text{rank} = 0+0+0+0 = 0$
$= 0$

$r_0(2) = n = 8$

$r_1(2) = 4$

$r_2(2) = 1$

$r_3(2) = 0$

$d_1(2) = r_0(2) - r_1(2) = 8 - 4 = 4$

$= \#$ of blocks of size ≥ 1

$d_2(2) = r_1(2) - r_2(2) = 4 - 1 = 3$

$= \#$ of blocks of size ≥ 2

$d_3(2) = r_2(2) - r_3(2) = 1 - 0 = 1$

$= \#$ of blocks ≥ 3

So these ranks completely determined the Jordan blocks in J.

The explain "uniqueness"

3/8/23

Important disclaimer on the computation of JCF:

We saw ranks of $(A-\lambda I)^m$ for all $\lambda \in \sigma(A)$ completely determine the JCF(A).

But it's one only approximation in practice, and rank determination can be numerically unstable:

e.g. $\begin{bmatrix} 1 & \varepsilon \\ 0 & 1 \end{bmatrix}$ rank $= 2$, eigenvalue $= 1$

$$\begin{bmatrix} 1 & \varepsilon \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & \varepsilon \\ 0 & 0 \end{bmatrix}$$

theoretically when $\varepsilon \neq 0$ rank $= 1$

practically if $\varepsilon \ll 1$ rank ≈ 0 by computer.

As a consequence, computation of JCF is virtually impossible, unless exact arithmetic can be used (symbolic)

However, JCF is <u>invaluable</u> in matrix theory.

Applications of JCF

(1) Role of eigenvalues in typical dynamical system that involve matrices.

input

$$x_0 \longrightarrow \boxed{D.S.} \longrightarrow x(t), \quad t \geq 0$$

Assymptotic behavior $x(t) \xrightarrow{t \to \infty} ?$ depends on the eigenvalues of the D.S.

Simple case

Ubiquitos dynamical system

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}, \quad x'(t) = \begin{bmatrix} x_1'(t) \\ x_2'(t) \\ \vdots \\ x_n'(t) \end{bmatrix}$$

$$A = [a_{ij}] \in M_n$$

$$\boxed{x'(t) = A\,x(t)}, \quad t \geq 0$$

(i.e. derivatives of input entries depend linearly on the entries of the state vector.)

" vector linear diff. equation "

Initial condition : $x(0) = x_0 \in \mathbb{C}^n$.

<u>Scalar case</u> $f'(t) = a f(t)$, $f: \mathbb{C} \to \mathbb{C}$.
differential

solution: $f(t) = e^{at}$

<u>Vector / matrix case</u> $x'(t) = A x(t)$, $A \in M_n$

Define <u>matrix exponential</u> e^{tA}, $t \geq 0$

$$e^{tA} := I + tA + \frac{t^2}{2!}A^2 + \cdots$$
$$= \sum_{k=0}^{\infty} \frac{(tA)^k}{k!}$$

This function exists (series converges), $\forall t$:

$$\|e^{tA}\| = \left\| \sum_{k=0}^{\infty} \frac{(tA)^k}{k!} \right\|$$

matrix norm

$$\leq \sum_{k=0}^{\infty} \frac{|t|^k \|A\|^k}{k!} \quad \text{(triangle \& submultipl.)}$$

$$= e^{t\|A\|} < \infty \quad \text{(converges?)}$$

To solve $x'(t) = Ax(t)$; $x(0) = x_0$

$$x(t) = e^{tA} x_0, \quad t \geq 0$$

because $(e^{tA} x_0)' = A e^{tA} x_0$. (easy)

How do the eigenvalues enter the discussion?

Say $A = SJS^{-1}$, $J = JCF(A)$

Change of variables

$$y(t) = S^{-1}x(t)$$

Then

$$y'(t) = S^{-1}x'(t) = S^{-1}Ax(t)$$

$$= \underbrace{S^{-1}AS}\,\underbrace{S^{-1}x(t)}$$

$$= Jy(t)$$

$\rightarrow$ Linear differential system with coefficient matrix $=$ Jordan matrix.

Let's for simplicity assume $J = J_n(\lambda)$

$$= \begin{bmatrix} \lambda & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix} \in M_n$$

<u>Then</u>

$\longrightarrow$

$$y' = Jy \quad \begin{cases} y_1'(t) = \lambda y_1(t) + y_2(t) \\ y_2'(t) = \lambda y_2(t) + y_3(t) \\ \vdots \\ y_{n-1}'(t) = \lambda y_{n-1}(t) + y_n(t) \\ y_n'(t) = \lambda y_n(t) \end{cases}$$

Solve this system backwards:

$$y_n(t) = y_n(0)\, e^{\lambda t}$$

$$y_{n-1}(t) = y_n(0)\, t e^{\lambda t} + y_{n-1}(0)\, e^{\lambda t}$$

$$\vdots$$

In general

$$y_k(t) = \underbrace{e^{\lambda t}} \cdot q_k(t)$$

where $q_k(t)$ = polynomial in t of degree $\leq n-k$

Easy to see that the term $e^{\lambda t}$ overpowers asymptotically $q_k(t)$ $(t \to \infty)$. So the behavior of

each $y_k(t)$

i.e. $y(t)$

i.e. $x(t) = Sy(t)$

depends on the eigenvalue(s) λ.

E.g. $|e^{\lambda t}| \xrightarrow{t\to\infty} \infty$ if $\lambda \in \mathbb{C}$, $\operatorname{Re}\lambda > 0$

$\xrightarrow{t\to\infty} 0$ if $\operatorname{Re}\lambda < 0$

oscillate if $\operatorname{Re}\lambda = 0$

<u>Definition</u> We call $A \in M_n$ a <u>stable</u> matrix if $\operatorname{Re}\lambda < 0$ for all $\lambda \in \sigma(A)$.

In such case, $x(t) = e^{tA} x_0 \xrightarrow{t\to\infty} 0$ for all $x_0 \in \mathbb{C}^n$.

("0" is a ~~sink~~ stable equilibrium)

It turns out that

discrete dynamical systems involve powers of matrices, & powers behave based on the eigenvalues too!

$$A^k \xrightarrow{k\to\infty} 0 \quad \text{iff} \quad \rho(A) < 1$$

spectral radius

We'll study this next.

$$\boxed{\rho(A) = \max\{|\lambda| : \lambda \in \sigma(A)\}}$$

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

A^k

3/10/23

<u>Midterm</u> : Correction on due date posted:

Due March 20, 5pm

#1 Find in notes, guess eigenvectors for parts $\underbrace{\text{(b) \& (c)}}_{\text{Brauer's Theorem.}}$

#2 Very basic.

#3 Part (c) : a bit of though of how to construct generic nilpotent matrices

#4 Part (b) : Follow hint. von Neumann's Lemma.

#5 (a) - (d) straightforward, (e) think!

Proceed with applications of JCF:

(2) We know that

$$P_A(t) = P_{A^T}(t)$$

for all $A \in M_n(\mathbb{R})$. That is, A & A^T are co-spectral. Are they similar?

YES: $A \sim A^T$ for all $A \in M_n(\mathbb{R})$.

<u>Proof</u>

$$A \sim JCF(A) = \bigoplus_{i=1}^{k} J_{n_i}(\lambda_{n_i})$$

To show $A \sim A^T$, it suffices to show that every Jordan block is similar to its transpose.

$$A^T \sim \begin{bmatrix} J_{n_1}(\lambda_1) & & 0 \\ & \ddots & \\ 0 & & J_{n_k}(\lambda_k) \end{bmatrix}^T = \begin{bmatrix} J_{n_1}(\lambda_1)^T & & \\ & \ddots & \\ & & J_{n_k}(\lambda_k)^T \end{bmatrix}$$

$$\sim \begin{bmatrix} J_{n_1}(\lambda_1) & & \\ & \ddots & \\ & & J_{n_k}(\lambda_k) \end{bmatrix} \sim A$$

However,

$$\underbrace{\begin{bmatrix} 0 & & 1 \\ & \iddots & \\ 1 & & 0 \end{bmatrix}}_{\substack{S \\ \| \\ S^{-1}}} \underbrace{\begin{bmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix}}_{J_k(\lambda)} \underbrace{\begin{bmatrix} 0 & & 1 \\ & \iddots & \\ 1 & & 0 \end{bmatrix}}_{S}$$

$$\overset{\text{carry}}{\underset{\text{out}}{=}} \begin{bmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix} = J_k(\lambda)^T$$

i.e. $\quad J_k(\lambda) \sim J_k(\lambda)^T$

and result follows. ∎

<u>Next</u> application of JCF to

$$\lim_{k \to \infty} A^k$$